

ULYANA DUPLETSA - CV

PhD in Astroparticle Physics

✉ ulyana.dupletsa@oeaw.ac.at — 🌐 Website — 🐙 GitHub — 🆔 ORCID

ABOUT ME

Astroparticle physicist working on **gravitational-wave (GW) cosmology** and **next-generation detector science**. My research focuses on standard-siren cosmology, compact-binary populations, Fisher-matrix forecasting, Bayesian parameter estimation, and multi-messenger astrophysics. I apply and develop statistical and computational methods to quantify how current and future GW observatories, especially **LIGO-Virgo-KAGRA (LVK)** and the **Einstein Telescope (ET)**, can constrain the expansion history of the Universe and the astrophysics of compact binaries.

CURRENT POSITION

Postdoctoral Research Fellow **Mar 2025-present**
Marietta Blau Institute (MBI), Austrian Academy of Sciences — Vienna, Austria

EDUCATION

PhD in Astroparticle Physics **Nov 2020-Feb 2025**
Gran Sasso Science Institute — L'Aquila, Italy
• PhD Thesis on “*Cosmic Chirps: Extracting Cosmological and Astrophysical Information from Compact Binary Mergers*”, defended cum Laude on **16th May 2025**, [link to thesis]
• Supervisor: Prof. Jan Harms; Co-Supervisors: Prof. Marica Branchesi, Dr. Riccardo Murgia

Post-MSc Research Activity **2020**
University of Milano Bicocca — Milan, Italy
• Research project with Prof. Alberto Sesana on supermassive black-hole binary evolution, modeling merger delays, triple black-hole systems, and PTA-relevant gravitational-wave backgrounds; GitHub 🐙

Master's degree in Theoretical Physics - 110/110 cum Laude **2017-2019**
University of Milano Bicocca — Milan, Italy
• Thesis on “*Thermodynamic Aspects of AdS_4 Black Holes in $N = 2$ Gauged Supergravity*” supervised by Prof. Alberto Zaffaroni

Bachelor's Degree in Physics - 110/110 cum Laude **2013-2017**
The University of Milano Bicocca — Milan, Italy
• Thesis on “*Supermassive Binary Black Holes and their Dynamics in Galactic Nuclei*” supervised by Prof. Monica Colpi

PUBLICATION LIST (SELECTED)

- “Radio sirens: inferring H_0 with binary black holes and neutral hydrogen in the era of the Einstein Telescope and the SKA Observatory”
U. Dupletsa, S. Mastrogiovanni, M. Spinelli, T. Ronconi, M. Schulz, R. Murgia, J. Harms, T. Baker, M. Calabrese, C. Carbone, S. Cunningham, I. Harrison, K. Leyde, D. Nanadoungar-Lacroze; *submitted to MNRAS*, arXiv:2605.12606 [astro-ph.CO] [astro-ph.HE]; Zenodo dataset, <https://doi.org/10.5281/zenodo.20582389>; GitHub 🐙
[Lead author.]
- “Validating Prior-informed Fisher-matrix Analyses against GWTC Data”
U. Dupletsa, J. Harms, Ken K. Y. Ng, J. Tissino, F. Santoliquido, A. Cozzumbo; *Physical Review D*, Vol. 111 (2025), 2, 024036,
DOI:10.1103/PhysRevD.111.024036, arXiv:2404.16103 [gr-qc], GitHub 🐙
[Lead author.]
- “GWFish: A simulation software to evaluate parameter-estimation capabilities of gravitational-wave detector networks”
U. Dupletsa, J. Harms, B. Banerjee, M. Branchesi, B. Goncharov, A. Maselli, A. C. S. Oliveira, S. Ronchini, J. Tissino; *Astronomy and Computing*, Vol. 42 (2023), DOI:10.1016/j.ascom.2022.100671, arXiv:2205.02499 [gr-qc], GitHub 🐙
[Lead author.]
- “GWTC-4.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation”
LIGO-Virgo-KAGRA Collaboration; arXiv:2509.04348 [astro-ph.CO], accepted in *APJ Letters*;
[Major analysis contribution with the icarogw pipeline, contribution to figure and table production and to paper writing.]
- “The Science of the Einstein Telescope”
A. Abac et al. incl. **U. Dupletsa**; JCAP 03 (2026) 081, arXiv:2503.12263 [gr-qc]
[Major contribution to the editorial and writing team of the Common Tools Division chapter, and part of the writing team for the Multi-Messenger Observations Division chapter.]

6. “Blinded Mock Data Challenge for Gravitational-Wave Cosmology-I: Assessing the Robustness of Methods Using Binary Black Holes Mass Spectrum”
A. Agarwal, **U. Dupletsa**, K. Leyde, S. Mukherjee, B. Revenu, J. E. Rivera, A. E. Romano, M. R. Sah, S. Vallejo-Pena, A. Avendano, F. Beirnaert, G. Dalya, M. C. Espitia, C. Karathanasis, S. Moreno-Gonzalez, L. Quiceno, F. Stachurski, J. Garcia-Bellido, R. Gray, N. Tamanini, C. Turski; *The Astrophysical Journal*, Vol. 987 (2025), DOI:10.3847/1538-4357/adda3a, arXiv:2412.14244 [astro-ph.CO], GitHub 
[Main analyst with the icarogw pipeline of the results presented in the paper, major figure and table production, and contribution to paper writing.]
7. “Prospects for optical detections from binary neutron star mergers with the next-generation multi-messenger observatories”
E. Loffredo, N. Hazra, **U. Dupletsa**, M. Branchesi, S. Ronchini, F. Santoliquido, A. Perego, B. Banerjee, S. Bisero, G. Ricigliano, S. Vergani, I. Andreoni, M. Cantiello, J. Harms, M. Mapelli, G. Oganesyan; *Astronomy & Astrophysics*, Vol. 697 (2025) A36, DOI:10.1051/0004-6361/202452863, arXiv:2411.02342 [astro-ph.HE]; Zenodo dataset <https://doi.org/10.5281/zenodo.13850416>
[One of the corresponding authors; GW-analysis lead: production and interpretation of the BNS results; contribution to figures and writing.]
8. “Science with the Einstein Telescope: a comparison of different designs”
M. Branchesi, M. Maggiore et al. incl. **U. Dupletsa**; *JCAP*, Vol. 07 (2023) 068, DOI:10.1088/1475-7516/2023/07/068, arXiv:2303.15923 [gr-qc] [astro-ph.CO] [astro-ph.HE]
[Production of all BNS results, cross-check of BBH results.]

ROLE IN LVK

LIGO-Virgo-KAGRA (LVK) Collaboration

CBC Cosmology Group member

- Main analyst and paper writing team member for the Mock Data Challenge short-author Collaboration paper using icarogw;
- Analysis with icarogw and the paper writing team member for the GWTC-4.0 Cosmology Collaboration paper;
- Analysis team member for the GWTC-5.0 Cosmology Collaboration paper;
- Coordinator of icarogw analyses for the GWTC-6.0 Cosmology Collaboration and paper writing team member;
- Internal reviewer for the GWTC-5.0 Rates & Population Collaboration paper figure scripts.

ROLE IN ET

Einstein Telescope (ET) Collaboration

Observational Science Board (OSB) member

- Member of Common Tools, Multi-Messenger Observations and Cosmology Divisions;
- Major contributor to ET design-comparison and cost-benefit studies through population forecasts with GWFish, resulting in two Collaboration-wide publications: Branchesi et al. 2023, and Abac et al. 2026;
- OSB liaison for ETO Task Force detector-layout studies; major contribution to workshops and science-performance assessments:

ETO Task Force Meeting (I) [\[link\]](#)

18-20 Feb 2025, *[online]*

ETO Task Force Meeting (II) [\[link\]](#)

18-20 Mar 2025, *Nikhef, Amsterdam*

ETO Task Force Meeting (III) [\[link\]](#)

5-7 May 2025, *CERN, Switzerland*

- Co-coordinator (together with Dr. Francesco Iacovelli) of the ET optical-layout science paper; currently under internal review.

INVITED TALKS & SEMINARS

Meeting of the Einstein Telescope Site Selection Group 1

11 May, 2026

[online monthly meeting]

- Talk on: “Assessing the Impact of Instrumental Requirements on the Scientific Performance of the Einstein Telescope”

Multiple Systems across Scales: First early-career researcher workshop of CRC1831 “Multiplicity of Stars and Planets” [\[link\]](#)

27-28 Apr, 2026

Heidelberg, Germany

- Talk on: “Standard siren cosmology: current status and prospects for the Einstein Telescope”

Seminar at Sapienza [\[link\]](#)

4 Mar, 2026

Rome, Italy

- Seminar on: “The present and future of GW cosmology”

Second International Conference on the Physics of the Two Infinities

17-21 Nov, 2025

[\[link\]](#)

Tokyo, Japan

- Talk on behalf of the ET Collaboration on: “The Einstein Telescope: challenges and opportunities of a third-generation detector”

CONFERENCES & WORKSHOPS

1st Sardinian GW science mini-workshop [link] <i>Cagliari, Italy</i> • Panelist in: “Cosmology and Black Hole Physics with GWs and 21cm intensity mapping”	22-26 Sep, 2025
LXVI Congresso nazionale della Società Astronomica Italiana [link] <i>Firenze, Italy</i> • Talk on: “Le sirene delle onde gravitazionali: calibrazione di un nuovo metodo per stimare H_0 ”	3-6 Jun, 2025
GraSP24 - Gravity Shape Pisa 2024 [link] <i>Pisa, Italy</i> • Talk on: “Forecasting detection and parameter estimation capabilities for the Einstein Telescope”	23-25 Oct, 2024
Seminar at MBI [link] <i>Vienna, Austria</i> • Seminar on: “Forecasting detection and parameter estimation capabilities for the Einstein Telescope”	10 Sep, 2024
The Seventeenth Marcel Grossmann Meeting [link] <i>Pescara, Italy</i> • Talk on “How to do multi-messenger forecasts in the Einstein Telescope era: addressing present and future challenges”	7-12 Jul, 2024
Seminar at Cagliari University [link] <i>Sardinia, Italy</i> • Seminar on: “ET forecasts on cosmological growth of BH masses”	23 Apr, 2024
ET: Scienza e Tecnologia in Italia [link] <i>Assisi, Italy</i> • Panelist in “Science and data analysis challenges”	20-23 Feb, 2024
GWADW - Gravitational Wave Advanced Detector Workshop [link] <i>Isola d’Elba, Italy</i> • Talk on: “Forecasting the Detection and Parameter Estimation Capabilities for different ET Designs”	22-26 May, 2023
XIII ET Symposium [link] <i>Cagliari, Italy</i> • Talk on: “Fisher Information Matrix for ET Forecasts: How Informative is it?”	8-12 May, 2023
LIGO-Virgo-KAGRA Collaboration meeting [link] <i>Pisa, Italy</i>	9-13 Mar, 2026
Einstein Telescope science workshop for early career researchers [link] <i>Sapienza University - Rome, Italy</i> (Co-organizer of the workshop)	18-20 Feb, 2026
2025 SKAO General Science Meeting [link] <i>Görlitz, Germany</i> • Talk on: “Mapping the Universe’s Expansion with Gravitational Waves and Neutral Hydrogen”	16-20 Jun, 2025
XV ET Symposium [link] <i>Bologna, Italy</i> • Plenary talk (shared with Dr. Mikhail Korobko) on: “OSB work for the ETO Task Force”	26-30 May, 2025
LIGO-Virgo-KAGRA Collaboration Meeting [link] <i>[online]</i> • Talk (shared with Gregoire Pierra) on: “Latest results from the GWTC-4 cosmology paper”	24-27 Mar, 2025
1st GSSI-JHU MAECI Meeting on Gravitational Wave Astrophysics [link] <i>L’Aquila, Italy</i> • Talk on: “Dark sirens cosmology”	16-18 Dec, 2024
AHEAD2024 2nd General Meeting [link] <i>Rome, Italy</i> • Talk on: “Parameter estimation catalogs for binary neutron star mergers detected with next-generation gravitational wave detectors”	26-27 Nov, 2024
LIGO-Virgo-KAGRA Collaboration Meeting [link] <i>Barcelona, Spain</i> • Talk (shared with Suvodip Mukherjee) on: “Updates on Mock Data Challenge - Analysis with icarogw”	23-26 Sep, 2024
ModIC 2024 - Model-Independent Cosmology with gravitational waves, large-scale structure, and high-energy surveys [link] <i>IFPU Trieste, Italy</i> • Talk on: “Dark Sirens Cosmology with neutral hydrogen intensity mapping”	13-17 May, 2024

XIV ET Symposium [link] <i>Maastricht, Netherlands</i>	6-10 May, 2024
• Talk on: “Enhancing Fisher Matrix Results with Physically Motivated Priors”	
Second Cosmic Explorer Symposium [link] <i>[online]</i>	23-25 Apr, 2024
• Short talk (1 minute 1 slide) on: “Enhancing Fisher Matrix Results with Physically Motivated Priors”	
Linking Advances in our Understanding of Theoretical Astrophysics and Relativity to Observations (LAUTARO) [link] <i>Milano-Bicocca, Italy</i>	17-19 Apr, 2024
• Talk on: “Enhancing Fisher Matrix Results with Physically Motivated Priors”	
LIGO-Virgo-KAGRA Collaboration Meeting [link] <i>[online]</i>	11-14 Mar, 2023
• Talk (shared with Suvodip Mukherjee) on: “MDC - Status and Prospects”	
LIGO-Virgo-KAGRA Collaboration Meeting [link] <i>Toyama, Japan</i>	11-14 Sept, 2023
• Talk (shared with Suvodip Mukherjee) on: “Mock Data Challenge - Analysis with icarogw”	
Amaldi 15 [link] <i>[online]</i>	17-21 Jul, 2023
• Talk on: “Forecasting detection and parameter estimation capabilities for different ET designs in a multi-messenger context”	
Gravitational-wave populations: what’s next? Workshop [link] <i>Milan, Italy</i>	10-14 Jul, 2023
• Short talk on: “GWFish - a Fisher Matrix Software”	
LIGO-Virgo-KAGRA Collaboration Meeting <i>Cardiff</i>	12-16 Sep, 2022
• Poster on: “GWFish - A simulation software to evaluate parameter-estimation capabilities of gravitational-wave detector networks”	
XII ET Symposium - The birth of the ET Collaboration [link] <i>Budapest, Hungary</i>	7-8 Jun, 2022

PHD SCHOOLS

IWR School on Machine Learning for Fundamental Physics [link] <i>Heidelberg, Germany</i>	15 - 19 Sep, 2025
Computing and High Performance Computing in Astronomy and Astrophysics [link] <i>Bologna, Italy</i>	24 Jun-5 Jul, 2024
International School for Subnuclear Physics 2024 - News from the four interactions [link] “Ettore Majorana Foundation” - Erice, Italy	14-23 Jun, 2024
• Winner of <i>New Talents: Giazotto prize</i> 🏆	
• Prize for <i>Best Presentation in Theoretical Physics</i> 🏆	
Scientific Communication in Astronomy School [link] <i>Bertinoro, Italy</i>	2-6 Oct, 2023
• <i>Winner of the school prize (3rd place)</i> 🏆	
First EuCapt School in Cosmology [link] <i>Valencia, Spain</i>	18-22 Sep, 2023
Amaldi Research Center Summer School [link] <i>Paestum, Italy</i>	5-9 Sep, 2022
The Onassis Foundation Science Lectures in Physics: Gravitational Waves [link] <i>Heraklion, Crete</i>	25-29 Jul, 2022

TECHNICAL SKILLS

Programming	• Python and Bash (advanced); C/C++ and Mathematica (intermediate); MATLAB and SQL (basic)
Scientific software	• GWFish, Bilby, icarogw, L ^A T _E X, git/GitHub, HTCondor, SLURM
Methods	• Gravitational-wave cosmology; standard sirens; compact-binary populations; Fisher forecasts; Bayesian parameter estimation; detector-network performance studies
Languages	• Italian and Ukrainian (native); English (advanced)

VISITINGS

GSSI	29 Apr-16 May, 2026
<i>L'Aquila, Italy</i>	
Visiting on ACME funds (link), collaboration on multi-messenger astrophysics and cosmology with the Einstein Telescope	
Sapienza University	3-8 Mar, 2026
<i>Rome, Italy</i>	
Visiting Dr. Simone Mastrogiovanni's group, and seminar on: "<i>The present and future of GW cosmology</i>" (link)	
GSSI	18-23 May, 2025
<i>L'Aquila, Italy</i>	
Discussion and collaboration for the ET Task Force documents preparation	
University of Cagliari	22-26 Apr, 2024
<i>Cagliari, Italy</i>	
Visiting Dr. Riccardo Murgia and seminar on "<i>ET forecasts on cosmological growth of BH masses</i>"	
Johns Hopkins University	Oct-Nov, 2023
<i>Baltimore, United States</i>	
Collaboration with Prof. Emanuele Berti's group (mainly with Dr. Ken Ng) as part of the exchange program MAECI MUR organized by Andrea Maselli, resulted in the publication of Dupletsa et al. 2025	

FUNDING & GRANTS

ACME funds - Selected visitor	Oct 2025
<i>Second TNA call to Centers of Expertise (link)</i>	
Selected for a two-week visiting program at GSSI for a collaboration on multi-messenger astrophysics and ET cosmology.	
APPHEAL Project minigrant - Co-Investigator	2025
<i>Advanced Platform for Processing Heavy Elements Libraries</i>	
€20k project funded by INAF; contribution to proposal writing [PI: Dr. Eleonora Loffredo]	

REVIEWER

Referee for the Astrophysical Journal (ApJ), internal reviewer in LVK

LECTURES & TUTORIALS

"Gravitational Waves and their detection" master course at the Technical University of Vienna	27 May, 2026
<i>Vienna, Austria</i>	
One lecture on "<i>Gravitational wave cosmology</i>" [2 hours] as part of the course held by Gianluca Inguglia	
PhD school on "Gravitational Waves: from theory to detection" (link)	7-18 Jul, 2025
<i>Vienna, Austria</i>	
Hands-on session on "<i>Metrics and tools for assessing the performance of future gravitational wave detectors</i>"; GitHub 	
"Gravitational Waves and their detection" master course at the Technical University of Vienna	4 Jun, 2025
<i>Vienna, Austria</i>	
One lecture on "<i>Gravitational wave cosmology</i>" [2 hours] as part of the course held by Gianluca Inguglia	
ModIC 2024 - Model-Independent Cosmology with gravitational waves, large-scale structure, and high-energy surveys (link)	13-17 May, 2024
<i>IFPU Trieste, Italy</i>	
Hands-on session on GWFish with and without priors; GitHub 	
ET: Scienza e Tecnologia in Italia (link)	20-23 Feb, 2024
<i>Assisi, Italy</i>	
School lecture (in collaboration with Jacopo Tissino) on "<i>GWFish: Simulation of gravitational-wave detector networks with Fisher-matrix PE</i>"; GitHub 	
XIII ET Symposium (link)	8-12 May, 2023
<i>Cagliari, Italy</i>	
Tutorial session on "<i>GWFish - a Fisher Matrix Analysis Software</i>"; GitHub 	

ORGANIZATION OF SCIENTIFIC EVENTS

Co-organizer of the MBI Institutional Colloquia (link)	Jan 2026- present
<i>Vienna, Austria</i>	
In collaboration with Prof. Claude Amsler	

Next generation gravitational wave observations: mini-colloquia [link] <i>Graz, Austria</i> • SOC member	21 Sep, 2026
Einstein Telescope Science Workshop for Early Career Researchers [link] <i>Sapienza University of Rome, Italy</i> • SOC member	18-20 Feb, 2026
ABETE: Avalanche of Bright sources towards the ET Era, The near-future prospects and challenges of multi-messenger astronomy [link] <i>Seston, Italy</i> • SOC member	9-13 Mar, 2026
XIII ET Symposium [link] <i>Cagliari, Italy</i> • Co-organizer of the Fisher matrix hackathon session	8-12 May, 2023

OUTREACH ACTIVITIES

- Lange Nacht der Forschung* — *Vienna, Austria*
• Outreach event organized by the Austrian Academy of Sciences
- SHARPER* — *L'Aquila, Italy*
• Regular contributor to European Researchers' Night / SHARPER (2021–2024).
- IDeAS* — *L'Aquila, Italy*
• Member of IDeAS (IDeAS is partner of Museo Nivola of Organi, GSSI, GSSI Alumni, INFN Cagliari, University of Cagliari, and INAF-OAC, in the realization of the project "*On the edge of the eyelid*", winner of *Strategia Fotografica 2024* from *Ministero della Cultura*)
• Public seminar on "*Seeing the Unseen*", in collaboration with Matteo Schulz and Andrea Cozzumbo, on 30th January 2025
- Scienza in Bidda* — *Jerzu, Sardinia*
• Event organized by Dr. Riccardo Murgia in 2023, in correspondence with the ET Symposium in Cagliari; part of the outreach project to talk about the relevance of the Einstein Telescope in Sardinia
- Cosmic Explorers* — *L'Aquila, Italy*
• Series of lectures by Prof. Marica Branchesi to elementary school kids, contribution to organization of activities and experiments with the kids; three days in March, April, and June 2023
- PhTea Talks* — *L'Aquila, Italy*
• Co-organizer and co-founder of informal talks for PhD students at GSSI; delivered outreach talks in 2023 and 2024
- Premio Asimov* — *L'Aquila, Italy*
• Project organized by Prof. Francesco Vissani; member of Premio Asimov reviewing committee, 2021–2026.